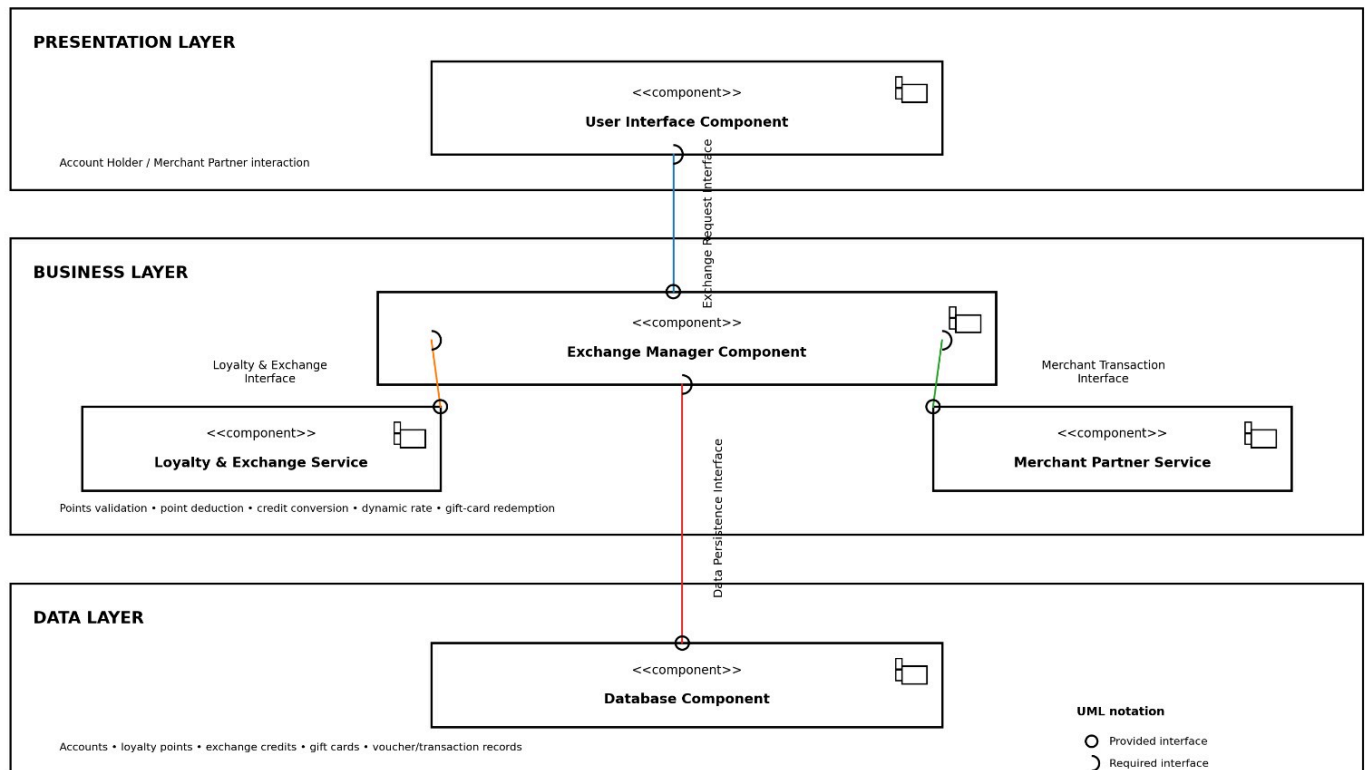


Digital Gift Card & Loyalty Point Exchange System

Layered Architecture — UML Component Diagram



LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Digital Gift Card & Loyalty Point Exchange

SRN: PES1UG24CS091 | Section: B | Domain: Retail, E-Commerce & Finance

Users: Account Holder, Merchant Partner

Architecture Selection

We chose Layered Architecture for the Digital Gift Card & Loyalty Point Exchange System.

Two Reasons

- 1. Clear separation of responsibilities:** The architecture separates the presentation, business and data concerns. The User Interface handles Account Holder and Merchant Partner interaction, while the Exchange Manager coordinates the core exchange workflow.
- 2. Centralized exchange rules:** The Exchange Manager can coordinate the Jira-defined workflow of validating loyalty points, deducting points, converting points to exchange credits using a dynamic conversion rate, checking eligible digital gift cards, and coordinating redemption. This keeps the business rules consistent across exchanges.

Security Advantage

The business layer provides a controlled boundary between users and stored transaction data. Exchange requests can be validated before points, credits, gift-card eligibility and voucher records are changed. This supports the security-related work represented in Jira, including voucher validation and preventing duplicate redemption.

Performance Benefit

The architecture keeps the main exchange flow direct: the User Interface communicates with the Exchange Manager, which coordinates the required service and database operations. Avoiding the additional network and operational overhead of a distributed microservices design makes the architecture suitable for responsive exchange processing and supports the system's strict performance objective.

Selected components: User Interface, Exchange Manager, Loyalty & Exchange Service, Merchant Partner Service, Database.

Required interfaces: Exchange Request, Loyalty & Exchange, Merchant Transaction, Data Persistence.